

ARCUS

FP&A Model Planning and Model Builder System

Developer Software Requirements Document

Financial Planning, Budgeting, Forecasting and Enterprise Model Management

Version 1.0 | July 2026

CONFIDENTIAL - Developer Specification

Document Control

Field	Requirement
Document name	Arcus FP&A Model Planning and Model Builder System - SRD
Document type	Developer Software Requirements Document
Product area	Financial Planning, Analysis, Budgeting, Forecasting and Enterprise Modelling
Primary users	FP&A administrators, finance managers, analysts, budget owners, CFOs, executives and auditors
Version	1.0
Status	Functional specification for design and development
Design direction	Modern institutional white interface optimised for 13.5-inch laptop screens, compact high-density planning tables, clear model state, blue and purple Arcus accents

Development principle

Users plan using business assumptions. The modelling engine translates those assumptions into financial outcomes. Model Builder defines the logic; Model Planning operates the plan.

1. Purpose

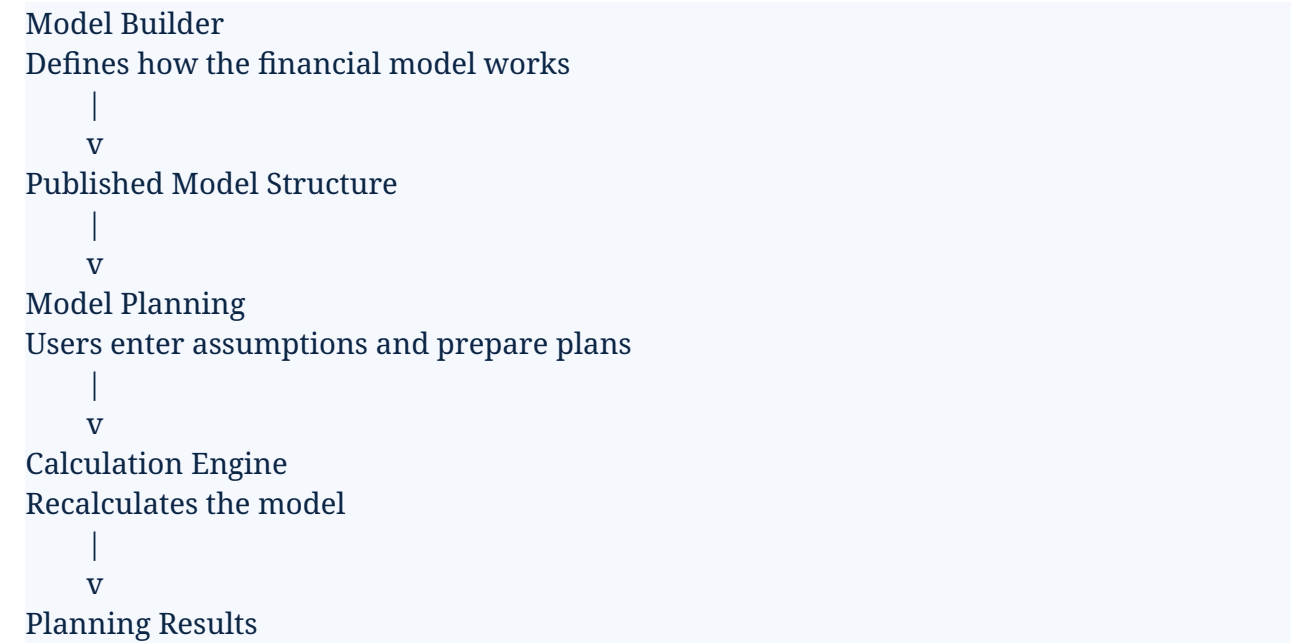
Developers must build **Model Planning** and **Model Builder** as two tightly connected components of the Arcus FP&A Forecasting System.

The two components solve different problems.

Model Planning is the business-facing planning workspace. Finance teams, departmental budget owners and executives use it to prepare budgets, update forecasts, compare scenarios, amend assumptions, review variances and move plans through approval workflows.

Model Builder is the technical modelling workspace. FP&A administrators and authorised power users use it to define the structure, calculations, dimensions, modules, line items, data mappings and dependencies that make Model Planning work.

The required operating relationship is:





The system must not treat Model Planning as a spreadsheet uploaded into Arcus.

The system must not treat Model Builder as a code editor.

Together, the two components must provide a controlled financial modelling engine that allows organisations to create, operate and govern complex budgets and forecasts without requiring normal planning users to understand the underlying technical architecture.

2. Plain-English Explanation for Developers

An FP&A team normally builds financial models to answer questions such as:

- What revenue do we expect next year?
- How many employees can we afford?
- What happens if sales fall by 15%?
- What happens if salaries increase by 8%?
- How much cash runway will remain?
- Which department is overspending?
- What is causing EBITDA to miss budget?
- What happens if the organisation opens a new branch?
- What happens if the USD/ZiG exchange rate changes?
- Can management compare the Budget, Forecast and Downside scenarios?
- Which numbers were manually entered and which were calculated?
- Who changed an assumption?
- Has the departmental plan been approved?

The actual calculations may be complicated.

For example:

Units Sold
x
Average Selling Price
=
Gross Revenue

Then:

Gross Revenue
-
Discounts
=
Net Revenue

Then:

Net Revenue

-

Cost of Goods Sold

=

Gross Profit

Then:

Gross Profit

-

Payroll

-

Marketing

-

Operating Expenses

=

EBITDA

A normal planning user should not have to rebuild these formulas.

The **Model Builder** defines the calculations once.

The **Model Planning** workspace then allows users to change authorised drivers such as:

Units Sold

Price

Headcount

Salary Increase

Inflation

FX Rate

Marketing Spend

Capex

Customer Growth

Churn

The system recalculates all dependent financial values automatically.

3. Product Architecture

Developers must separate the system into six logical layers.

1. Data Source Layer

|

v

2. Dimensional Model Layer

|

v

3. Calculation and Dependency Engine

|

v

4. Model Builder

|

v

5. Model Planning

|
v

6. Workflow, Reporting and Audit Layer

3.1 Data Source Layer

Sources may include:

- Accounting General Ledger.
- Payroll.
- Procurement.
- Fixed Assets.
- CRM.
- Sales systems.
- Stock or inventory systems.
- Loan systems.
- Portfolio systems.
- Banking data.
- CSV.
- Excel.
- API integrations.
- Manual assumptions.

The source layer provides actual data or external planning drivers.

3.2 Dimensional Model Layer

The dimensional layer defines how data is organised.

Example dimensions:

Time
Department
Entity
Product
Region
Customer Segment
Account
Cost Centre
Employee
Project
Currency
Version
Scenario

The calculation engine must use these dimensions to determine the exact intersection at which a value exists.

Example:

Revenue
January 2026
Harare Branch
Product A
Budget 2026
Base Case
USD

is a different value from:

Revenue
January 2026
Bulawayo Branch
Product A
Budget 2026
Base Case
USD

The system must not store planning values as unstructured spreadsheet coordinates.

3.3 Calculation and Dependency Engine

The calculation engine must:

- Parse formulas.
- Resolve references.
- Calculate line items.
- Detect circular dependencies.
- Maintain dependency graphs.
- Calculate summaries.
- Calculate dimensional intersections.
- Perform scenario calculations.
- Perform version calculations.
- Execute allocations.
- Recalculate affected downstream values.
- Record calculation errors.

3.4 Model Builder

Model Builder defines:

What exists
+
How it is structured
+
How it is calculated
+
Where its data comes from

3.5 Model Planning

Model Planning controls:

What users assume
+
What users change
+
Which scenario they are planning
+
Which version they are working on
+
Who reviews the plan

3.6 Workflow, Reporting and Audit Layer

The system must provide:

- Planning workflows.
- Submission.
- Review.
- Approval.
- Commentary.
- Tasks.
- Variance analysis.
- Dashboards.
- Management reports.
- Financial statements.
- Audit logs.

4. Application Navigation

The FP&A system shall include:

```

FP&A
|-- Home
|-- Dashboards
|-- Reports
|-- Driver Library
|-- Model Planning
|   |-- Planning Workspace
|   |-- Scenario Comparison
|   |-- Department Plans
|   |-- Assumptions
|   |-- Variance Analysis
|   |-- Planning Commentary
|   `-- Planning History
|-- Model Builder
|   |-- Models
|   |-- Modules
|   |-- Line Items
|   |-- Dimensions
|   |-- Formula Editor
|   |-- Dependency Map
|   |-- Data Mapping
|   |-- Validations
|   `-- Publishing History
|-- Scenarios
|-- Data Hub
|-- Integrations
|-- Workflows
|-- Tasks
|-- Alerts
`-- Administration
  
```

5. Core Terminology

Developers must use the following definitions consistently.

5.1 Model

A model is the complete logical financial-planning structure.

Example:

FY2026 Corporate Financial Model

A model may contain:

- Revenue planning.
- Workforce planning.
- Operating expenses.
- Capital expenditure.
- Cash flow.
- Balance sheet.
- Profit and loss.
- KPI calculations.

5.2 Module

A module is a logical calculation area inside a model.

Examples:

Revenue Planning
Workforce Planning
Operating Expenses
Capital Expenditure
Working Capital
Cash Flow
Financial Statements

5.3 Line Item

A line item represents one business or financial measure.

Examples:

Units Sold
Average Selling Price
Revenue
Headcount
Average Salary
Payroll Expense
Marketing Spend
EBITDA
Cash Balance

5.4 Dimension

A dimension determines how a value is segmented.

Example:

Revenue
BY
Product
Region
Month
Scenario
Version

5.5 Driver

A driver is an input or assumption that influences another calculation.

Examples:

Volume Growth
Price Increase
Salary Increase
Inflation
Customer Churn
FX Rate

5.6 Version

A version represents a distinct planning dataset.

Examples:

FY2026 Budget
Forecast Q1
Forecast Q2
Forecast Q3
Board Approved Budget
Working Forecast

5.7 Scenario

A scenario represents a set of assumptions applied to a version.

Examples:

Base Case
Best Case
Downside
Severe Downside
Management Case

Version and scenario are not interchangeable.

Example:

Version: Forecast Q3
Scenario: Downside

is different from:

Version: FY2026 Budget
Scenario: Downside

5.8 Actual

Actual is data that has already occurred and been imported from a source system.

Example:

January 2026 Revenue Actual

Actual data must normally be read-only in Model Planning.

5.9 Forecast

Forecast represents expected future performance.

5.10 Plan

Plan is the complete dataset currently being prepared or approved.

6. User Roles

6.1 FP&A Administrator

May:

- Create models.
- Create modules.
- Create dimensions.
- Create line items.
- Define formulas.
- Configure data mappings.
- Validate models.
- Publish model versions.
- Manage model permissions.

This user works primarily in Model Builder.

6.2 FP&A Analyst

May:

- Update planning assumptions.
- Prepare forecasts.
- Analyse variances.
- Create scenarios.
- Add commentary.
- Review departmental plans.
- Run comparison analysis.

6.3 Finance Manager

May:

- Coordinate the planning cycle.
- Assign planning tasks.
- Review submissions.
- Reopen plans.
- Submit consolidated plans.
- Approve configured workflow stages.

6.4 Department Budget Owner

Examples:

- Sales Director.
- Marketing Manager.
- Head of Operations.
- Head of Technology.

May:

- View assigned planning sections.
- Update authorised inputs.
- Add commentary.
- Submit departmental plans.

May not:

- Change formulas.
- Create model dimensions.
- Change data mappings.
- Publish a model.

6.5 CFO

May:

- View consolidated plans.
- Compare scenarios.
- Review material variances.
- Approve plans.
- Approve major assumption changes.
- Approve final budget or forecast.

6.6 Executive or Board User

Read-only or controlled approval access.

May:

- View dashboards.
- Compare scenarios.
- Review approved planning assumptions.
- Review forecasts.
- Add comments where permitted.

6.7 Auditor

Read-only access to:

- Model versions.
- Formulas.
- Change history.
- Approvals.
- Data sources.
- Calculation history.
- Audit logs.

PART A - MODEL BUILDER

7. Model Builder Purpose

Developers must build Model Builder as the technical modelling environment for Arcus FP&A.

It must allow authorised users to create a financial model by defining:

Dimensions

|

v

Modules

|

v

Line Items

|

v

Formulas

|

v

Data Sources

|

v

Dependencies

|

v

Validations

|

v

Published Model

A user must be able to create complex models without writing backend application code.

However, the system must remain governed.

Model changes must be:

- Validated.
- Versioned.
- Tested.
- Audited.
- Published.

A change made in Model Builder must not silently alter an approved financial plan.

8. Model Builder Main Workflow

Create model

|

v

Configure model calendar and base currency

|

v

Create or select dimensions

|

v

Create modules

|

v

Configure module dimensionality

|

v

Create line items

|

v

Define input or calculated line-item type

|

v

Configure formulas

|

v

Map source data

|

v

Generate dependency graph

|

v

Run validation

|

v

Test calculations

|

v

Resolve errors and warnings

|

v

Publish model

|

v

Model becomes available in Model Planning

9. Model Creation

When creating a model, the system must request:

- Model name.
- Description.
- Financial year start month.
- Base currency.
- Planning frequency.
- Planning horizon.
- Historical data horizon.
- Default scenario.
- Default version.

- Model owner.
- Workspace.
- Access group.

Example:

Model Name:
FY2026 Financial Model

Financial Year Start:
January

Frequency:
Monthly

Historical Period:
January 2023 - December 2025

Planning Horizon:
January 2026 - December 2030

Base Currency:
USD

Supported planning frequencies:

Daily
Weekly
Monthly
Quarterly
Annual

Monthly should be the default FP&A frequency.

10. Dimension Management

10.1 Purpose

Dimensions provide the analytical structure of the model.

Developers must not hard-code Department, Region or Product directly into the financial tables.

They must be configurable dimensions.

10.2 Standard System Dimensions

The system must provide:

Time
Version
Scenario
Currency

These are controlled system dimensions.

10.3 User Dimensions

Users may create:

Department
 Cost Centre
 Product
 Region
 Entity
 Branch
 Customer Segment
 Employee
 Project
 Channel
 Account
 Asset Class

10.4 Dimension Hierarchy

Dimensions must support parent-child hierarchies.

Example:

```

Zimbabwe
|-- Harare
|   |-- Borrowdale
|   `-- CBD
`-- Bulawayo
  
```

Another example:

```

Operating Expenses
|-- Staff Costs
|   |-- Salaries
|   |-- Bonuses
|   `-- Benefits
|-- Marketing
`-- Administration
  
```

10.5 Aggregation

Parent values must normally aggregate child values.

Parent = SUM(Children)

Example:

```

Total Zimbabwe Revenue
=
Harare Revenue
+
Bulawayo Revenue
  
```

Supported summary methods:

SUM
 AVERAGE
 MIN
 MAX
 FIRST
 LAST
 COUNT
 WEIGHTED_AVERAGE
 NONE
 FORMULA

10.6 Dimension Member Controls

Each dimension member must store:

- Member ID.
- Code.
- Name.
- Parent.
- Display order.
- Start date.
- End date.
- Active status.
- Attributes.
- Source system.
- External key.

Inactive members must remain available for historical calculations.

11. Module Management

A model may contain multiple modules.

Example:

```
FY2026 Financial Model
|-- Revenue Planning
|-- Workforce
|-- Opex Planning
|-- Capex
|-- Working Capital
|-- Cash Flow
`-- Financial Statements
```

Each module must store:

- Module name.
- Description.
- Model.
- Module owner.
- Dimensions.
- Time scale.
- Version applicability.
- Scenario applicability.
- Calculation order.

- Status.

Module statuses:

DRAFT
VALID
INVALID
PUBLISHED
ARCHIVED

12. Module Dimensionality

Every module must explicitly define its dimensions.

Example:

Revenue Planning
Dimensions:
Time
Product
Region
Version
Scenario

Workforce may use:

Time
Department
Employee
Version
Scenario

The engine must therefore understand that Revenue may exist by Product and Region, while Payroll Expense may exist by Department and Employee.

The system must support references between modules with different dimensionality.

13. Line Item Builder

13.1 Required Line Item Types

The system must support:

INPUT
CALCULATED
IMPORTED
SYSTEM
REFERENCE
ALLOCATION
BOOLEAN
TEXT
DATE
LIST

13.2 Input Line Item

A user may enter planning data.

Example:

Units Sold
Type: INPUT

13.3 Calculated Line Item

The value is determined by a formula.

Example:

Revenue = Units Sold x Price

The user must not directly overwrite Revenue.

13.4 Imported Line Item

Populated from an external source.

Example:

Actual Payroll Expense
Source: Payroll

13.5 Reference Line Item

References a value from another module.

Example:

Headcount = Workforce.Headcount

13.6 Allocation Line Item

Calculated by an allocation rule.

Example:

Corporate IT Cost
allocated across departments
based on Headcount

14. Required Line Item Fields

Each line item must include:

- Line item ID.
- Module.
- Name.
- Description.
- Type.
- Data type.
- Format.
- Currency.
- Decimal places.
- Formula.
- Summary method.
- Dimensionality.
- Input permissions.
- Source mapping.
- Validation rule.

- Display order.
- Active status.

15. Formula Engine

The formula engine is one of the most critical components.

Developers must create a parser that supports references using readable business terminology.

Example:

```
[Units Sold] * [Price]
```

Cross-module reference:

```
[Workforce.Headcount] * [Workforce.Average Salary]
```

16. Required Formula Operators

Arithmetic:

```
+  
-  
*  
/  
^
```

Comparison:

```
=  
!=  
>  
<  
>=  
<=
```

Logical:

```
AND  
OR  
NOT
```

17. Required Formula Functions

At minimum:

```
IF()  
SUM()  
AVERAGE()  
MIN()  
MAX()  
ABS()  
ROUND()  
ROUNDUP()  
ROUNDDOWN()  
COUNT()  
COUNTIF()
```

SUMIF()
 IFERROR()
 ISBLANK()
 PREVIOUS()
 NEXT()
 OPENING()
 CLOSING()
 GROWTH()
 LAG()
 LEAD()
 MOVINGAVERAGE()
 YTD()
 QTD()
 MTD()
 CAGR()
 NPV()
 IRR()
 XIRR()

18. Formula Examples

18.1 Revenue

$[Revenue] = [Units\ Sold] * [Price]$

18.2 Discount

$[Discounts] = [Revenue] * [Discount\ %]$

18.3 Net Revenue

$[Net\ Revenue] = [Revenue] - [Discounts]$

18.4 Cost of Goods Sold

$[COGS] = [Units\ Sold] * [COGS\ per\ Unit]$

18.5 Gross Profit

$[Gross\ Profit] = [Net\ Revenue] - [COGS]$

18.6 Gross Margin

$Gross\ Margin = (Gross\ Profit / Net\ Revenue) * 100$

Formula:

IF(
 $[Net\ Revenue] = 0,$
 0,
 $[Gross\ Profit] / [Net\ Revenue]$
)

18.7 Payroll Expense

```
[Payroll Expense]
=
[Headcount]
*
[Average Fully Burdened Cost]
```

18.8 EBITDA

```
[EBITDA]
=
[Gross Profit]
-
[Payroll Expense]
-
[Marketing Spend]
-
[Other Operating Expenses]
```

18.9 Closing Cash

```
[Closing Cash]
=
[Opening Cash]
+
[Operating Cash Flow]
+
[Financing Cash Flow]
+
[Investing Cash Flow]
```

19. Time-Based Calculations

19.1 Opening and Closing Balances

Example:

```
January Closing Cash = 500,000
February Opening Cash = January Closing Cash
```

Formula:

```
[Opening Cash] = PREVIOUS([Closing Cash])
```

19.2 Growth

```
Growth_t = (Value_t - Value_t-1) / Value_t-1
```

Formula:

```
GROWTH([Revenue])
```

19.3 Year-to-Date

```
YTD([Revenue])
```

For June:

January Revenue

+

February Revenue

+

March Revenue

+

April Revenue

+

May Revenue

+

June Revenue

19.4 Moving Average

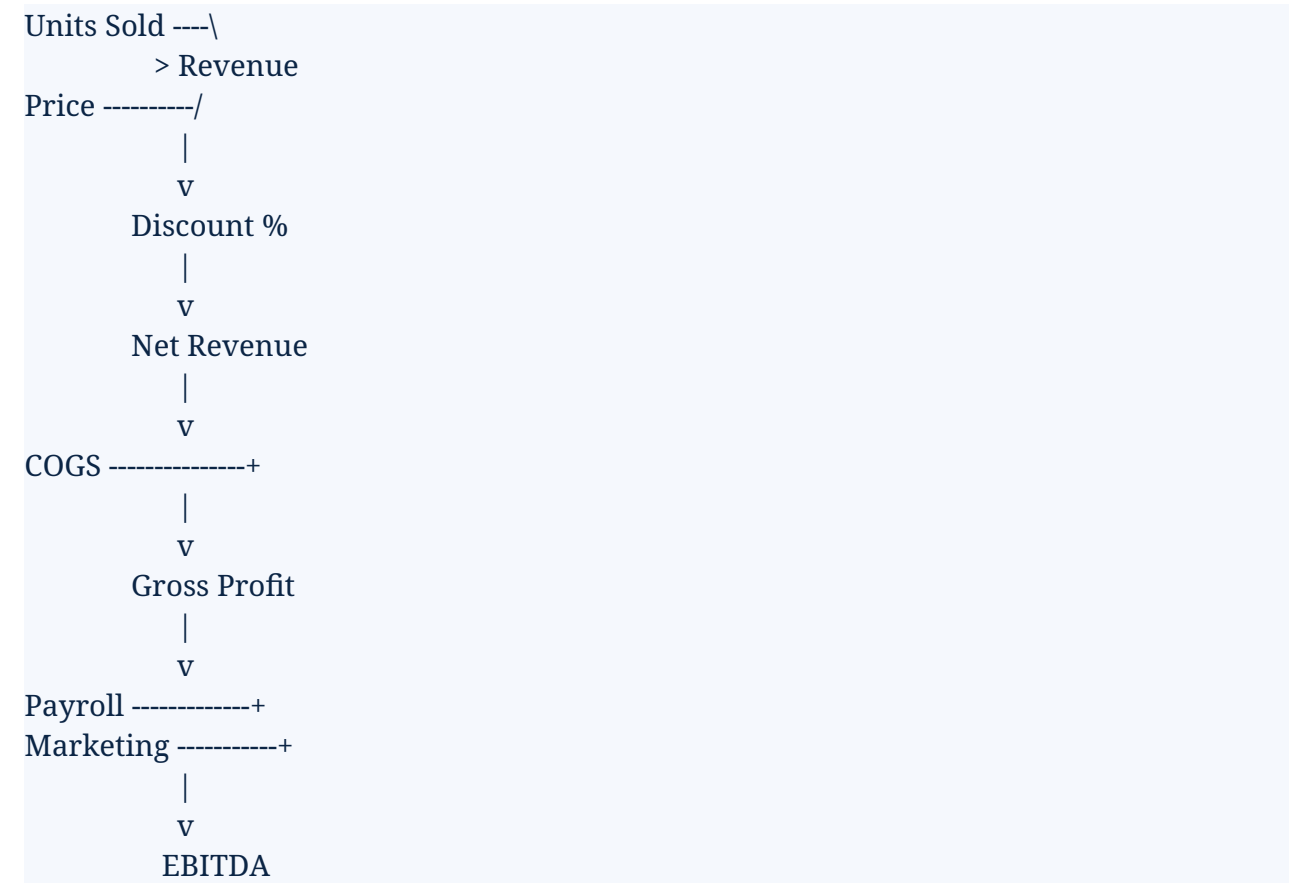
Example:

MOVINGAVERAGE([Revenue], 3)

returns the three-period average.

20. Dependency Engine

Every calculated line item creates dependencies.



The engine must build a **Directed Acyclic Graph (DAG)**.

20.1 Recalculation Logic

When Price changes:

```

Price
|
v
Revenue
|
v
Discount
|
v
Net Revenue
|
v
Gross Profit
|
v
EBITDA
|
v
Cash Flow
|
v
Closing Cash

```

Only affected downstream calculations should be recalculated.

The system should not recalculate the entire model where unnecessary.

20.2 Circular Reference Detection

Example:

```

Revenue = Price x Units
Units = Revenue / Price

```

This creates:

```

Revenue -> Units -> Revenue

```

The model must be invalid.

Validation error:

```

CIRCULAR_REFERENCE

Revenue Planning.Revenue
-> Revenue Planning.Units
-> Revenue Planning.Revenue

```

Publishing must be blocked.

21. Dependency Map UI

Model Builder must include a graphical dependency map.

Users must be able to:

- Select a module.
- Select a line item.

- View precedent calculations.
- View dependent calculations.
- Zoom.
- Filter external inputs.
- Highlight errors.
- Navigate directly to a related line item.

Legend:

Blue line Direct reference
 Dotted line Indirect reference
 Purple line External input
 Red line Invalid dependency

22. Data Mapping

Imported line items must map to external source fields.

Example:

GL_ACCOUNT_4000
 NetSuite GL
 ->
 Revenue

Another example:

HR_HEADCOUNT
 Payroll
 ->
 Headcount

Required mapping fields:

- Source system.
- Source object.
- Source field.
- Source account.
- Target model.
- Target module.
- Target line item.
- Dimension mapping.
- Transformation.
- Currency rule.
- Aggregation.
- Effective date.
- Mapping status.

23. Mapping Statuses

MAPPED
 PARTIALLY_MAPPED
 UNMAPPED
 INVALID
 DISABLED

The Model Builder dashboard must show a mapping completion percentage such as:

96% Mapped

A model may be published with configured non-critical warnings.

Critical unmapped values must block publishing.

24. Allocation Engine

The system must support cost allocations.

Example:

Corporate IT Expense = \$100,000

Allocate by departmental headcount:

Department	Headcount
Sales	40
Technology	30
Operations	20
Finance	10
Total	100

Sales allocation:

$100,000 \times (40 / 100) = 40,000$

General formula:

$\text{Allocation}_i = \text{Pool} \times (\text{Driver}_i / \text{SUM}(\text{Driver}))$

Supported allocation drivers:

- Headcount.
- Revenue.
- Floor area.
- Units sold.
- Asset value.
- Transaction count.
- Manual weight.
- Custom calculated line item.

25. Model Validation

The system must validate the model before publishing.

Validation categories:

FORMULA
DEPENDENCY
MAPPING
DIMENSION
DATA_TYPE
CURRENCY
TIME
SECURITY
PERFORMANCE

Severity:

ERROR
WARNING
INFO

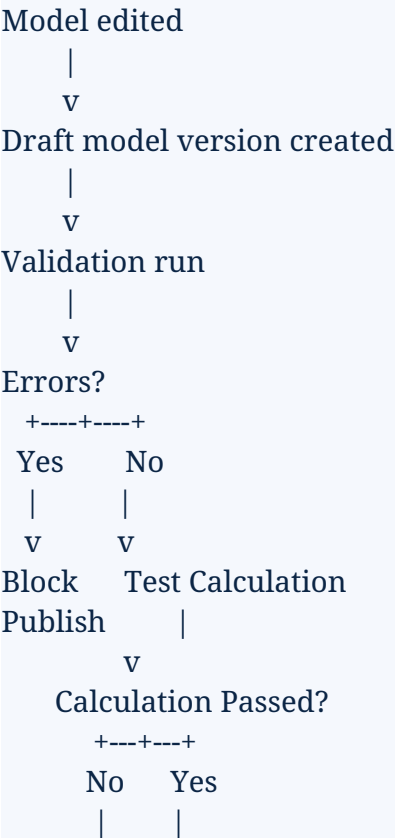
Example errors:

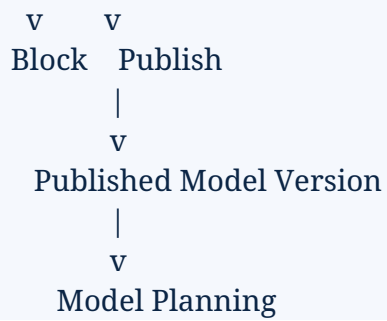
Circular reference detected.
Missing mandatory source mapping.
Formula references deleted line item.
Invalid dimension relationship.
Division by zero not handled.

Example warnings:

Driver has no forecast data after September.
Fallback exchange rate is being used.
Line item has not been updated in 12 months.

26. Model Publishing Workflow





A model publish record must include:

- Model version.
- Published by.
- Published date.
- Validation result.
- Model checksum.
- Modules changed.
- Line items changed.
- Formula changes.
- Mapping changes.

PART B - MODEL PLANNING

27. Model Planning Purpose

Model Planning is the operational planning environment for FP&A teams and budget owners.

Users must be able to:

- Select a planning cycle.
- Select a model version.
- Select a scenario.
- Enter authorised assumptions.
- View calculated outputs.
- Compare scenarios.
- Analyse variance.
- Comment on values.
- Assign planning tasks.
- Submit plans.
- Review plans.
- Approve plans.

The Model Planning workspace must hide unnecessary technical modelling complexity.

A department manager should see:

Marketing Spend
Headcount Plan
Campaign Volume
Customer Acquisition

They should not need to understand:

Calculation DAG
Dimension Intersection
Line Item Dependency
Module Dimensionality

28. Planning Cycle Setup

A planning cycle must contain:

- Planning cycle name.
- Financial year.
- Planning type.
- Source model.
- Source model version.
- Actuals cut-off period.
- Forecast start period.
- Planning horizon.
- Base scenario.
- Submission deadline.
- Approval workflow.
- Planning owner.

Planning types:

ANNUAL_BUDGET
 ROLLING_FORECAST
 QUARTERLY_FORECAST
 LONG_RANGE_PLAN
 STRATEGIC_PLAN
 REFORECAST

29. Budget Workflow

```

FP&A creates FY2027 Budget cycle
|
v
Selects published model
|
v
Actual FY2026 data imported
|
v
Prior budget copied where required
|
v
Planning assumptions created
|
v
Department plans opened
|
v
Budget owners enter drivers
|
v
System recalculates financial model
|
v
Department submits plan
|
v
Finance reviews
|
v
Comments and amendments
|
v
Plan resubmitted
|
v
FP&A consolidates plans
|
v
CFO reviews scenarios
|
  
```

```

    v
Final plan approved
    |
    v
Approved budget version locked

```

30. Rolling Forecast Workflow

```

Existing Forecast Q2
    |
    v
July actuals imported
    |
    v
July forecast replaced with actual
    |
    v
Forecast start moves to August
    |
    v
Future assumptions copied forward
    |
    v
FP&A reviews forecast drivers
    |
    v
Departments amend future periods
    |
    v
Model recalculates
    |
    v
Forecast Q3 submitted
    |
    v
Review and approval
    |
    v
Forecast Q3 locked

```

31. Actual and Forecast Cut-Over

The system must distinguish actual periods from forecast periods.

Example:

```

January - May = Actual
June - December = Forecast

```

Default rule:

If Period <= Actuals Cut-Off:
Value = Actual Source Data

If Period > Actuals Cut-Off:
Value = Planning Value

Users must not edit actual periods unless granted an exceptional adjustment permission.

32. Planning Grid

The planning grid is the primary Model Planning workspace.

Rows represent:

- Departments.
- Products.
- Accounts.
- Drivers.
- Line items.

Columns normally represent:

- Months.
- Quarters.
- Years.
- Scenarios.
- Versions.

Example:

Department	Jan	Feb	Mar	Apr	FY Total
Sales & Marketing	2,154	2,246	2,307	2,415	30,055
Product	1,876	1,924	1,987	2,043	25,453
Engineering	3,245	3,312	3,445	3,568	43,649

33. Cell Behaviour

Every planning-grid cell must have a defined status.

ACTUAL
INPUT
CALCULATED
IMPORTED
LOCKED
OVERRIDE
ERROR

Visual treatment must distinguish:

- Editable input.
- Calculated cell.
- Actual data.
- Locked value.
- Cell with comments.
- Validation issue.

The system must not rely only on colour. Icons or tooltips should reinforce status.

34. Cell-Level Audit

Clicking a value must allow users to view:

Value: 2,480,000
Line Item: Revenue
Period: May 2026
Department: Sales
Version: Forecast Q3
Scenario: Base Case

Calculation:
Units Sold x Price

Units Sold: 24,800
Price: 100

Result:
2,480,000

For an input:

Previous Value: 5.0%
New Value: 6.5%
Changed By: Tendai Moyo
Changed At: 12 May 2026 14:32
Reason: Revised supplier inflation assumption

35. Input Controls

Inputs may use:

- Numeric field.
- Percentage.
- Currency.
- Date.
- Boolean.
- Dropdown.
- Dimension selection.
- Text.

Validation examples:

Headcount ≥ 0
Discount % ≥ 0
Discount % ≤ 100
FX Rate > 0
Tax Rate ≥ 0

The system must reject invalid input before committing the value.

36. Spreading and Phasing

Users must be able to spread values across periods.

Example:

Annual Marketing Budget = \$1,200,000

Even spread:

\$100,000 per month

Supported methods:

EVEN
PRIOR_YEAR_PATTERN
CUSTOM_WEIGHT
WORKING_DAYS
SEASONAL_PROFILE
HISTORICAL_PATTERN
MANUAL

36.1 Even Spread

Monthly = Annual Value / 12

36.2 Weighted Spread

Period_i = Annual Value x Weight_i

where:

SUM(Weight_i) = 1

Example:

Jan 5%
Feb 5%
Mar 8%
...
Dec 15%

37. Driver-Based Planning

The system should encourage planning through drivers.

Example revenue planning:

Customers
x
Average Revenue per Customer
=
Revenue

Another example:

Sales Representatives
x
Deals per Representative
x
Average Deal Value
=
New Bookings

Another example:

```

Opening Headcount
+
New Hires
-
Exits
=
Closing Headcount

```

The Driver Library must maintain reusable planning drivers.

38. Scenario Management

Users must be able to create scenarios.

Example:

```

Base Case
Best Case
Downside

```

A scenario should normally inherit from a base scenario.

Example:

```

Downside
inherits
Base Case

```

Only changed assumptions need to be stored.

Example:

```

Base Case Revenue Growth = 8.7%
Downside Revenue Growth = -4.0%

```

Other assumptions may continue to inherit from Base Case.

39. Scenario Overlay Logic

The effective value is:

```

Scenario Override
if one exists

ELSE

Base Scenario Value

```

Formula:

```
EffectiveValue = OverrideValue if override exists; otherwise BaseValue
```

This reduces unnecessary data duplication.

40. Scenario Comparison

The Model Planning screen must support side-by-side scenario comparison.

Required columns:

Budget
Forecast
Best Case
Base Case
Downside
Variance to Budget

Metrics may include:

- Revenue.
- COGS.
- Gross Profit.
- Gross Margin.
- Opex.
- EBITDA.
- EBITDA Margin.
- Capex.
- Headcount.
- Cash Balance.

41. Variance Calculations

41.1 Absolute Variance

Variance = Actual or Forecast - Plan

41.2 Percentage Variance

Variance % = ((Actual - Plan) / ABS(Plan)) x 100

Where Plan = 0, the UI must show:

N/A

rather than an infinite percentage.

41.3 Favourable and Unfavourable Variance

The system must understand metric behaviour.

For Revenue:

Actual > Budget = Favourable

For Expense:

Actual > Budget = Unfavourable

Each line item must support:

variance_direction =
HIGHER_IS_FAVOURABLE
LOWER_IS_FAVOURABLE
NEUTRAL

42. Waterfall Variance Analysis

Users must be able to bridge one financial result to another.

Example:

Bridge item	Value
Budget Revenue	118.2M
Price Impact	+10.2M
Volume Impact	+5.1M
Mix Impact	-3.2M
Other Income	+2.6M
Opex Impact	-2.5M
Forecast Revenue	128.4M

For revenue:

$$\text{Revenue Variance} = \text{Price Impact} + \text{Volume Impact} + \text{Mix Impact} + \text{Other Impact}$$

The calculation engine should allow configurable variance bridges.

43. Assumption Management

The planning workspace must include an Assumptions panel.

Driver	FY2025 Actual	FY2026 Plan	Change
Volume Growth	12.5%	9.8%	-2.7 pp
Price Change	3.2%	2.5%	-0.7 pp
Headcount	512	568	+56
Inflation	3.4%	2.6%	-0.8 pp
FX USD/EUR	1.08	1.12	+0.04

Each assumption must have:

- Owner.
- Source.
- Effective period.
- Scenario.
- Version.
- Confidence.
- Commentary.
- Approval status.

44. Planning Commentary

Users must be able to comment at:

- Model level.
- Module level.
- Line item level.
- Dimension member.
- Cell level.
- Variance.
- Scenario.

Example:

@Tariro Moyo

Can you confirm whether the Q4 increase in marketing spend includes the product-launch campaign?

Comments must support:

- User mentions.
- Replies.
- Attachments.
- Resolution status.
- Time stamp.

45. Planning Tasks

A planning comment or workflow event may create a task.

Example:

Review Marketing Plan
Owner: Rutendo Nyathi
Due: 16 May

Task statuses:

NOT_STARTED
IN_PROGRESS
BLOCKED
COMPLETED
CANCELLED
OVERDUE

46. Planning Workflow

Default workflow:

```
DRAFT
|
v
SUBMITTED
|
v
UNDER_REVIEW
|
v
APPROVED
```

Alternative states:

RETURNED
REOPENED
REJECTED
LOCKED

Example:

Department Budget Owner

|

v Submit

Finance Manager

|

v Review

FP&A Team

|

v Consolidate

CFO

|

v Approve

Approved Plan

47. Submission Controls

Before submitting a plan, the system must check:

- Mandatory assumptions completed.
- No blocking validation errors.
- Required commentary provided.
- Assigned tasks completed.
- Material variances explained.
- Plan owner identified.

Failed submission example:

Submission blocked.

3 requirements remain:

- Headcount assumption missing for Operations.
- Marketing variance of 32% requires commentary.
- FX assumption has not been reviewed.

48. Material Variance Commentary

A configurable rule must require commentary where:

$\text{ABS}(\text{Variance } \%) \geq \text{Threshold}$

Example:

Threshold = 10%

Where marketing expense is 18% above plan, commentary is mandatory.

49. Model Planning Dashboard

The main Model Planning screen must show:

Header

- Model version.
- Planning cycle.
- Scenario tabs.
- Compare button.

- Actions menu.

KPI Cards

- Revenue.
- Gross Margin.
- EBITDA.
- Cash Runway.
- Headcount.
- Variance to Plan.

Main Planning Grid

Detailed monthly planning values.

Lower Panels

- Revenue versus expense trend.
- Driver assumptions.
- Variance waterfall.
- Scenario assumptions.

Right Panel

- Comments.
- Approvals.
- Activity.
- Tasks.

Workflow Footer

Draft -> Submitted -> Under Review -> Approved

50. Cash Runway Formula

Example:

$\text{Cash Runway} = \text{Current Cash} / \text{Average Monthly Net Cash Burn}$

Where monthly cash burn is positive.

If:

Current Cash = \$14,600,000

Average Monthly Burn = \$1,000,000

Then:

Cash Runway = 14.6 months

If the organisation is cash-generative, display:

Cash Generative

instead of a negative runway.

51. Headcount Planning Logic

Basic formula:

Closing Headcount

=

Opening Headcount

+

New Hires

-

Exits

Payroll:

Average Headcount

=

(Opening Headcount + Closing Headcount) / 2

Payroll Expense

=

Average Headcount

x

Average Fully Burdened Monthly Cost

Where detailed employee planning is used, employee-level calculations may replace the average-headcount method.

52. Integrated Model Flow

DATA HUB

Actual GL, Payroll, CRM, Operational Data

|

v

MODEL BUILDER

Dimensions

Modules

Line Items

Formulas

Mappings

|

v

PUBLISHED MODEL

|

v

MODEL PLANNING

Budget Drivers

Forecast Assumptions

Scenario Overrides

|

v

CALCULATION ENGINE

Dependency Resolution

Allocations

Currency Conversion

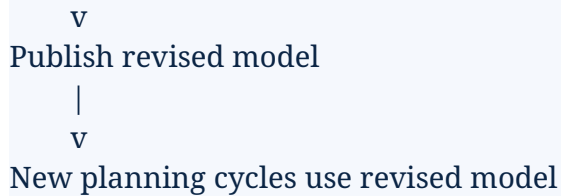
Time Calculations

|

v
 PLANNING OUTPUT
 Revenue
 Expenses
 EBITDA
 Cash Flow
 Balance Sheet
 KPIs
 |
 v
 WORKFLOW
 Submit
 Review
 Comment
 Approve
 |
 v
 APPROVED VERSION
 |
 v
 REPORTING AND DASHBOARDS

53. Model Change Flow

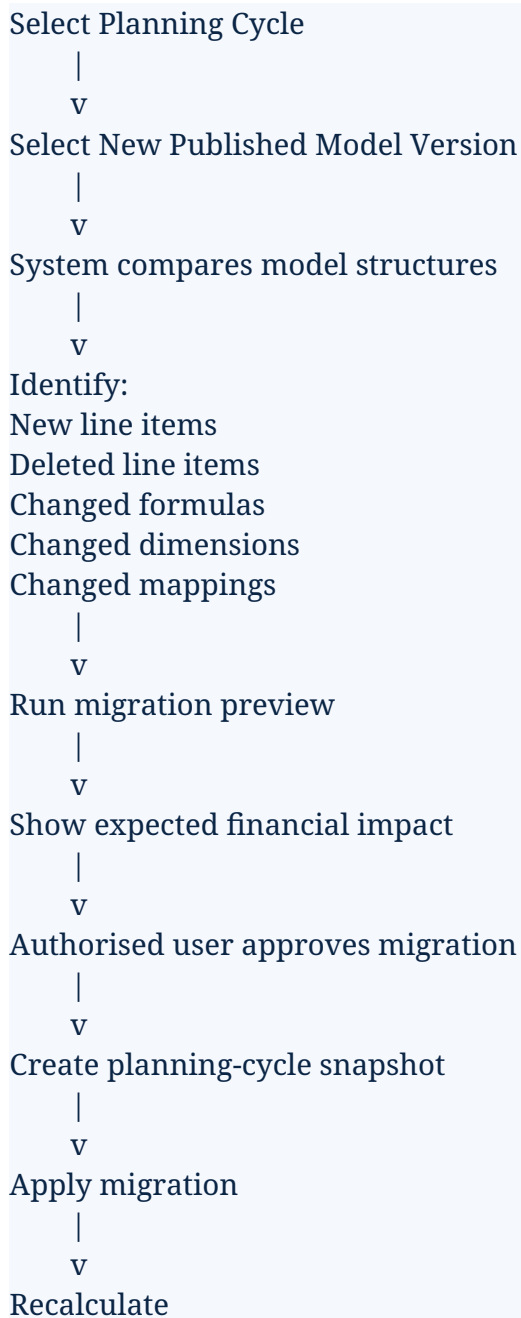
FP&A Administrator identifies modelling requirement
 |
 v
 Open Model Builder
 |
 v
 Create draft model revision
 |
 v
 Change module / line item / formula
 |
 v
 Dependency graph automatically updates
 |
 v
 Run validation
 |
 v
 Run test calculation
 |
 v
 Compare new results against published version
 |
 v
 Review impact analysis
 |



An active approved planning version must not automatically recalculate using a newly published model unless an authorised migration is performed.

54. Model Migration

Where a new model version is applied to an existing planning cycle:



55. User Stories

55.1 Create a Revenue Formula

User Story

As an FP&A Administrator, I want to define Revenue as Units Sold multiplied by Price so that planning users only need to update the business drivers.

Acceptance Criteria

- Units Sold and Price exist.
- Revenue is configured as calculated.
- Formula validates.
- Dependency map displays both inputs.
- Revenue recalculates after either input changes.
- Planning users cannot directly edit Revenue.

55.2 Prepare a Department Budget

User Story

As a Department Manager, I want to enter my department's headcount and expense assumptions so that Finance can consolidate the annual budget.

Acceptance Criteria

- User sees assigned department only.
- Editable cells are clearly indicated.
- Calculated cells are read-only.
- Invalid values are rejected.
- User can add commentary.
- User can submit the plan.

55.3 Compare Scenarios

User Story

As a CFO, I want to compare Base Case, Best Case and Downside forecasts so that I can understand the financial effect of different assumptions.

Acceptance Criteria

- Scenarios appear side by side.
- Revenue, EBITDA and Cash are calculated independently.
- Scenario assumptions are visible.
- Variance to Budget is calculated.
- Users can open the source driver.

55.4 Trace an EBITDA Calculation

User Story

As an FP&A Analyst, I want to trace how EBITDA was calculated so that I can investigate an unexpected forecast result.

Acceptance Criteria

- User selects EBITDA.
- Formula is displayed.
- Precedents are shown.
- Dependency graph appears.
- Current precedent values are displayed.

- User can drill into each precedent.

55.5 Detect Circular Formula

User Story

As an FP&A Administrator, I want the system to identify circular formula references before publishing so that invalid models cannot enter planning.

Acceptance Criteria

- Circular dependency is detected.
- Validation shows all affected line items.
- Dependency map highlights the cycle.
- Publishing is blocked.

55.6 Rolling Forecast Cut-Off

User Story

As an FP&A Manager, I want completed forecast periods to be replaced with actual values so that the forecast always reflects the latest financial results.

Acceptance Criteria

- Actual period is imported.
- Period becomes read-only.
- Prior forecast value remains in history.
- Forecast start period moves forward.
- Future assumptions remain available.

56. Database Design

56.1 Core Tables

```
fpa_models
fpa_model_versions
fpa_modules
fpa_line_items
fpa_dimensions
fpa_dimension_members
fpa_module_dimensions
fpa_line_item_formulas
fpa_formula_dependencies
fpa_data_mappings
fpa_allocation_rules
fpa_planning_cycles
fpa_plan_versions
fpa_scenarios
fpa_scenario_overrides
fpa_planning_values
fpa_actual_values
fpa_cell_comments
fpa_planning_tasks
fpa_workflow_instances
fpa_workflow_actions
fpa_validation_results
```

```
fpa_model_publish_logs
fpa_model_migrations
fpa_calculation_jobs
fpa_audit_logs
```

57. fpa_models

```
CREATE TABLE fpa_models (
  model_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  model_name VARCHAR(255) NOT NULL,
  description TEXT,
  financial_year_start_month INTEGER NOT NULL,
  base_currency VARCHAR(3) NOT NULL,
  default_frequency VARCHAR(50) NOT NULL,
  planning_horizon_start DATE NOT NULL,
  planning_horizon_end DATE NOT NULL,
  model_owner_id UUID NOT NULL,
  status VARCHAR(50) DEFAULT 'DRAFT',
  created_by UUID NOT NULL,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
```

58. fpa_modules

```
CREATE TABLE fpa_modules (
  module_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  model_id UUID NOT NULL REFERENCES fpa_models(model_id),
  module_name VARCHAR(255) NOT NULL,
  description TEXT,
  calculation_order INTEGER,
  time_scale VARCHAR(50),
  status VARCHAR(50) DEFAULT 'DRAFT',
  module_owner_id UUID,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);
```

59. fpa_line_items

```
CREATE TABLE fpa_line_items (
  line_item_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  module_id UUID NOT NULL REFERENCES fpa_modules(module_id),
  line_item_name VARCHAR(255) NOT NULL,
  description TEXT,
  line_item_type VARCHAR(50) NOT NULL,
  data_type VARCHAR(50) NOT NULL,
  format_type VARCHAR(50),
  currency_code VARCHAR(3),
  decimal_places INTEGER DEFAULT 2,
  summary_method VARCHAR(50) DEFAULT 'SUM',
  variance_direction VARCHAR(50) DEFAULT 'NEUTRAL',
```

```

display_order INTEGER,
is_active BOOLEAN DEFAULT TRUE,
created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

60. fpa_line_item_formulas

```

CREATE TABLE fpa_line_item_formulas (
  formula_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  line_item_id UUID NOT NULL
  REFERENCES fpa_line_items(line_item_id),
  model_version_id UUID NOT NULL,
  formula_expression TEXT NOT NULL,
  parsed_formula JSONB,
  validation_status VARCHAR(50) DEFAULT 'PENDING',
  created_by UUID NOT NULL,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

61. fpa_planning_values

```

CREATE TABLE fpa_planning_values (
  planning_value_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  planning_cycle_id UUID NOT NULL,
  plan_version_id UUID NOT NULL,
  scenario_id UUID NOT NULL,
  line_item_id UUID NOT NULL,
  dimension_intersection JSONB NOT NULL,
  period_id UUID NOT NULL,
  numeric_value NUMERIC(30,10),
  text_value TEXT,
  boolean_value BOOLEAN,
  value_source VARCHAR(50) NOT NULL,
  cell_status VARCHAR(50) NOT NULL,
  entered_by UUID,
  entered_at TIMESTAMP WITH TIME ZONE,
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

62. Formula Dependency Table

```

CREATE TABLE fpa_formula_dependencies (
  dependency_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  model_version_id UUID NOT NULL,
  source_line_item_id UUID NOT NULL,
  dependent_line_item_id UUID NOT NULL,
  dependency_type VARCHAR(50) NOT NULL,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
  UNIQUE (
    model_version_id,
    source_line_item_id,

```

```

        dependent_line_item_id
    )
);

```

63. API Requirements

63.1 Model APIs

```

POST /api/v1/fpa/models
GET /api/v1/fpa/models
GET /api/v1/fpa/models/{model_id}
PATCH /api/v1/fpa/models/{model_id}
POST /api/v1/fpa/models/{model_id}/create-version
POST /api/v1/fpa/models/{model_id}/validate
POST /api/v1/fpa/models/{model_id}/publish

```

63.2 Model Builder APIs

```

POST /api/v1/fpa/models/{model_id}/modules
GET /api/v1/fpa/models/{model_id}/modules
POST /api/v1/fpa/modules/{module_id}/line-items
PATCH /api/v1/fpa/line-items/{line_item_id}
POST /api/v1/fpa/line-items/{line_item_id}/formula
POST /api/v1/fpa/formulas/{formula_id}/validate
GET /api/v1/fpa/line-items/{line_item_id}/dependencies
GET /api/v1/fpa/line-items/{line_item_id}/impact

```

63.3 Dimension APIs

```

POST /api/v1/fpa/dimensions
GET /api/v1/fpa/dimensions
POST /api/v1/fpa/dimensions/{dimension_id}/members
PATCH /api/v1/fpa/dimension-members/{member_id}
GET /api/v1/fpa/dimensions/{dimension_id}/hierarchy

```

63.4 Planning APIs

```

POST /api/v1/fpa/planning-cycles
GET /api/v1/fpa/planning-cycles
GET /api/v1/fpa/planning-cycles/{cycle_id}
GET /api/v1/fpa/planning/{cycle_id}/grid
POST /api/v1/fpa/planning/{cycle_id}/values
POST /api/v1/fpa/planning/{cycle_id}/spread
POST /api/v1/fpa/planning/{cycle_id}/calculate
POST /api/v1/fpa/planning/{cycle_id}/submit
POST /api/v1/fpa/planning/{cycle_id}/approve
POST /api/v1/fpa/planning/{cycle_id}/return

```

63.5 Scenario APIs

```

POST /api/v1/fpa/scenarios
GET /api/v1/fpa/scenarios
POST /api/v1/fpa/scenarios/{scenario_id}/override
GET /api/v1/fpa/scenarios/compare
POST /api/v1/fpa/scenarios/{scenario_id}/calculate

```

64. Calculation Engine Guardrails

64.1 Never Use UI Values as the Source of Truth

The server must recalculate formulas.

Do not trust calculated values sent from the browser.

Incorrect:

Browser sends:
Revenue = 1,500,000

Correct:

Browser sends:
Units Sold = 15,000
Price = 100

Server calculates:
Revenue = 1,500,000

64.2 Never Store Calculated Cells as Manual Inputs

Calculated line items must remain calculated.

Manual override requires a separate override record.

64.3 Prevent Circular Dependencies

The formula graph must be validated before saving a published model.

64.4 Preserve Historical Plans

Do not recalculate approved plans using a changed formula silently.

Approved plan versions must remain reproducible.

64.5 Never Overwrite Actuals With Forecast Values

Actual and plan data must remain separate.

64.6 Do Not Approve an Invalid Model

Publishing requires:

Critical Validation Errors = 0

64.7 Recalculate on the Server

Formula evaluation, scenario overlay, summary calculations and allocations must execute in the backend calculation engine.

65. Security Requirements

Permissions must be enforced at:

- Model.
- Model version.
- Module.
- Line item.
- Dimension.

- Dimension member.
- Planning cycle.
- Scenario.
- Department.
- Entity.

Example:

A Marketing Manager may have:

READ:

Total Company

WRITE:

Marketing Department

A Regional Manager may have:

WRITE:

Zimbabwe Region

NO ACCESS:

Zambia Region

API-level permission checks are mandatory.

66. Performance Requirements

The system must be designed for large planning models.

Target behaviour:

- Planning grid initial load: under 3 seconds for normal views.
- Cell input acknowledgement: under 1 second.
- Small dependency recalculation: under 2 seconds.
- Large model calculation: asynchronous job with progress status.
- Model validation: asynchronous where required.
- Dependency graph load: under 3 seconds for filtered views.

Developers should use:

- Sparse value storage.
- Dependency-directed recalculation.
- Calculation caching.
- Batch writes.
- Bulk imports.
- Database indexing.
- Asynchronous calculation jobs.

67. Audit Requirements

The audit system must record:

- User.
- Action.
- Model.
- Module.
- Line item.

- Planning cycle.
- Scenario.
- Version.
- Old value.
- New value.
- Formula change.
- Timestamp.
- Reason.
- Source IP.
- Session.

Material audit events include:

MODEL_CREATED
 MODULE_CREATED
 LINE_ITEM_CREATED
 FORMULA_CHANGED
 MAPPING_CHANGED
 MODEL_VALIDATED
 MODEL_PUBLISHED
 PLANNING_VALUE_CHANGED
 SCENARIO_OVERRIDE_CREATED
 PLAN_SUBMITTED
 PLAN_RETURNED
 PLAN_APPROVED
 PLAN_REOPENED

68. UI Requirements - Model Planning

The interface must follow the modern Arcus FP&A design language.

Required characteristics:

- White background.
- Premium institutional appearance.
- Compact left navigation.
- Blue and purple accent states.
- High-density but readable planning tables.
- Light grey borders.
- Rounded cards.
- Numerical alignment.
- Compact toolbar.
- Comments and task side panels.
- Designed for a 13.5-inch laptop screen.

The Model Planning screen must include:

Model Version
 Planning Cycle
 Scenario Tabs
 Compare
 Actions

KPI cards:

Revenue
Gross Margin
EBITDA
Cash Runway
Headcount

Main areas:

Scenario Comparison / Planning Grid
Variance Waterfall
Scenario Assumptions
Comments
Approvals
Activity
Tasks

69. UI Requirements - Model Builder

The Model Builder must visually communicate that the user is configuring the system.

Required layout:

Left Panel
Modules
Dimensions

Centre
Line Item Builder
Formula Grid
Dependency Map

Right Panel
Properties
Formula Editor
References

Alternative detailed builder screen:

Top:
Model
Workspace
Validation Status
Test Calculation
Change History
Publish

Main:
Line Item Builder

Lower:
Formula Trace
Data Mapping
Audit Log
Exceptions

Validation Summary

Impact Analysis

Users must be able to open a line item and immediately understand:

- What it is.
- Whether it is input or calculated.
- Its formula.
- Its dimensions.
- Its source.
- What depends on it.
- Whether it is valid.

70. Acceptance Criteria

The system shall be accepted when:

1. Users can create financial planning models.
2. Users can create configurable dimensions.
3. Dimensions support parent-child hierarchy.
4. Users can create modules with different dimensionality.
5. Users can create input and calculated line items.
6. Calculated line items support readable formulas.
7. The formula engine correctly calculates arithmetic expressions.
8. Cross-module references work.
9. Time-based functions work.
10. Summary methods work.
11. Allocation rules work.
12. The system automatically creates formula dependencies.
13. Circular references are detected.
14. Circular references block model publishing.
15. Users can view formula precedents and dependents.
16. Dependency maps show calculation relationships.
17. External source fields can be mapped to line items.
18. Invalid mappings appear in validation results.
19. Users can validate a model.
20. Users can run test calculations.
21. Users can publish a valid model.
22. Published model versions remain historically reproducible.
23. Planning cycles can use published models.
24. Actual periods are separated from forecast periods.
25. Planning users can enter authorised assumptions.
26. Planning users cannot edit calculated line items.
27. Planning values are validated.
28. Input changes recalculate dependent financial values.
29. Users can spread annual values across periods.
30. Users can create planning scenarios.
31. Scenario values may inherit from a base scenario.
32. Users can compare Budget, Forecast and scenarios.
33. Absolute and percentage variances are calculated.
34. Favourable and unfavourable variance logic is configurable.
35. Users can review variance waterfalls.
36. Users can comment at cell and planning levels.

37. Users can assign planning tasks.
38. Plans can move through Draft, Submitted, Review and Approval.
39. Submission validation blocks incomplete plans.
40. Material variance commentary can be mandatory.
41. Approved versions can be locked.
42. Model changes do not silently alter approved plans.
43. Existing planning cycles can use controlled model migration.
44. All model changes are audit logged.
45. All planning-value changes are audit logged.
46. Security is enforced at model and dimensional levels.
47. The Model Planning interface supports planning, scenario comparison and collaboration.
48. The Model Builder interface supports line items, formulas, mappings, dependencies and validation.
49. The system supports dependency-directed recalculation.
50. The final FP&A model produces reproducible financial outputs.

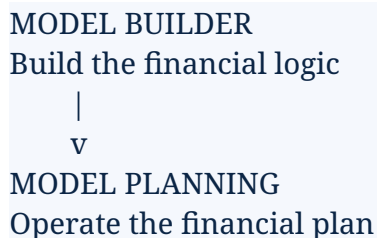
71. Final Product Definition

The completed Arcus **Model Planning and Model Builder System** must provide one controlled modelling and planning environment for professional FP&A teams.

Model Builder must allow authorised finance power users to define how the organisation's financial model works. They must be able to configure dimensions, modules, line items, formulas, allocations, data mappings and calculation dependencies without requiring changes to the core Arcus application code.

Model Planning must allow finance teams and budget owners to use the published model to prepare budgets, forecasts and scenarios through business assumptions rather than technical formulas.

The relationship between both components must remain clear:



The final system must allow FP&A teams to understand **what changed, why it changed, who changed it, how the financial outcome was calculated and what happens under alternative business assumptions.**

It must behave as a governed enterprise financial modelling engine rather than a collection of independent spreadsheets.

Appendix A - UI Inspiration

The following high-fidelity UI inspiration screens form part of the design brief. They use the Arcus white institutional interface, compact 13.5-inch screen layout, blue and purple accents, high-density financial tables and business-readable modelling controls.

A.1 Model Planning - Planning Workspace

UI inspiration: planning cycle and model version selectors, scenario tabs, KPI cards, high-density planning grid, trends, driver assumptions, comments, tasks and workflow status.

A.2 Model Planning - Scenario Comparison

UI inspiration: side-by-side Budget, Forecast, Best Case, Base Case and Downside comparison, variance-to-plan table, waterfall analysis, scenario assumptions, comments and approval status.

A.3 Model Builder - Modules, Line Items and Dependency Map

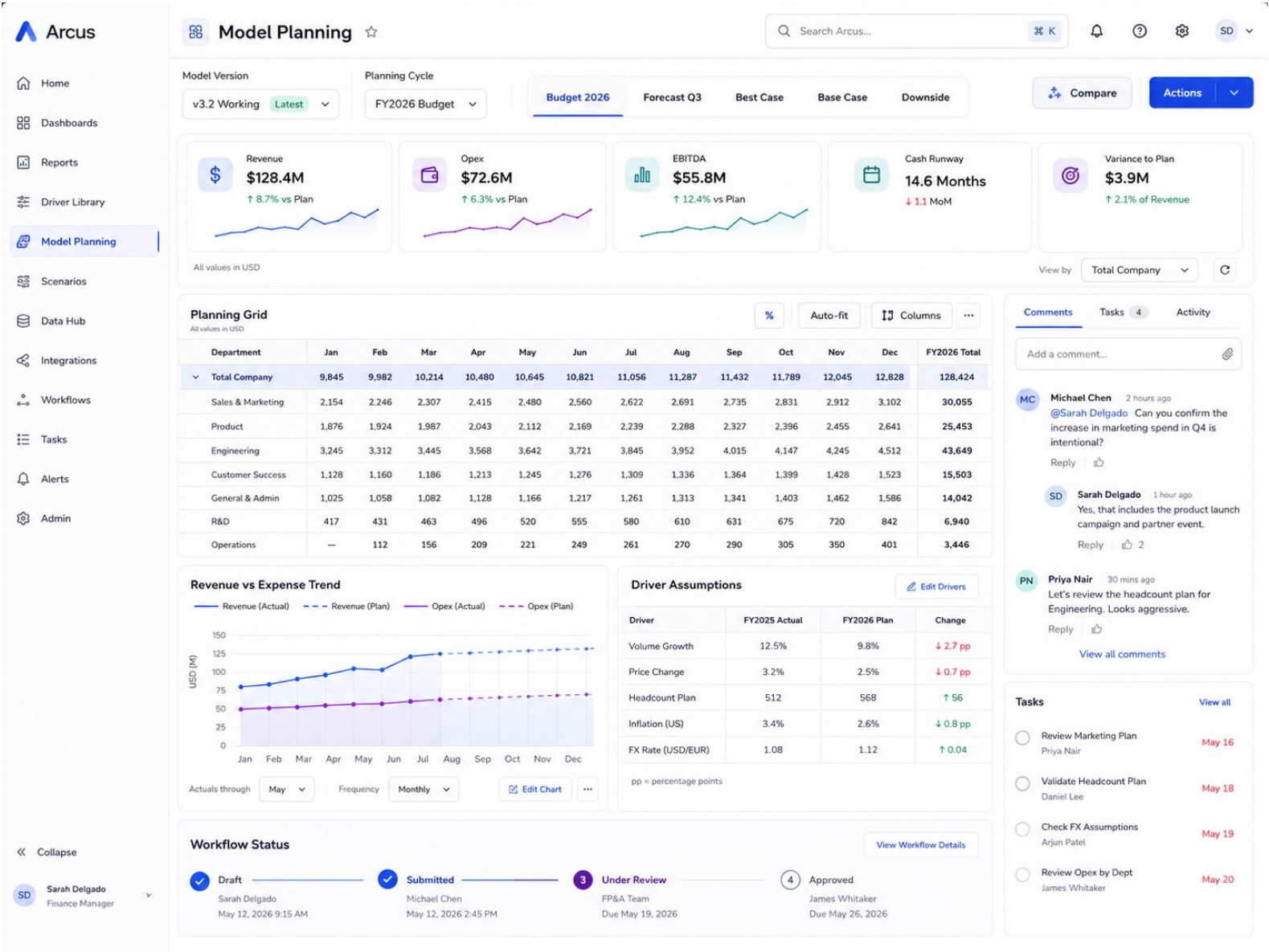
UI inspiration: modules and dimensions on the left, line-item formula grid in the centre, properties and formula editor on the right, with dependency mapping below.

A.4 Model Builder - Formula Trace, Mapping and Validation

UI inspiration: line-item builder, formula trace, source-to-target data mapping, model audit log, exceptions, validation summary and impact analysis.

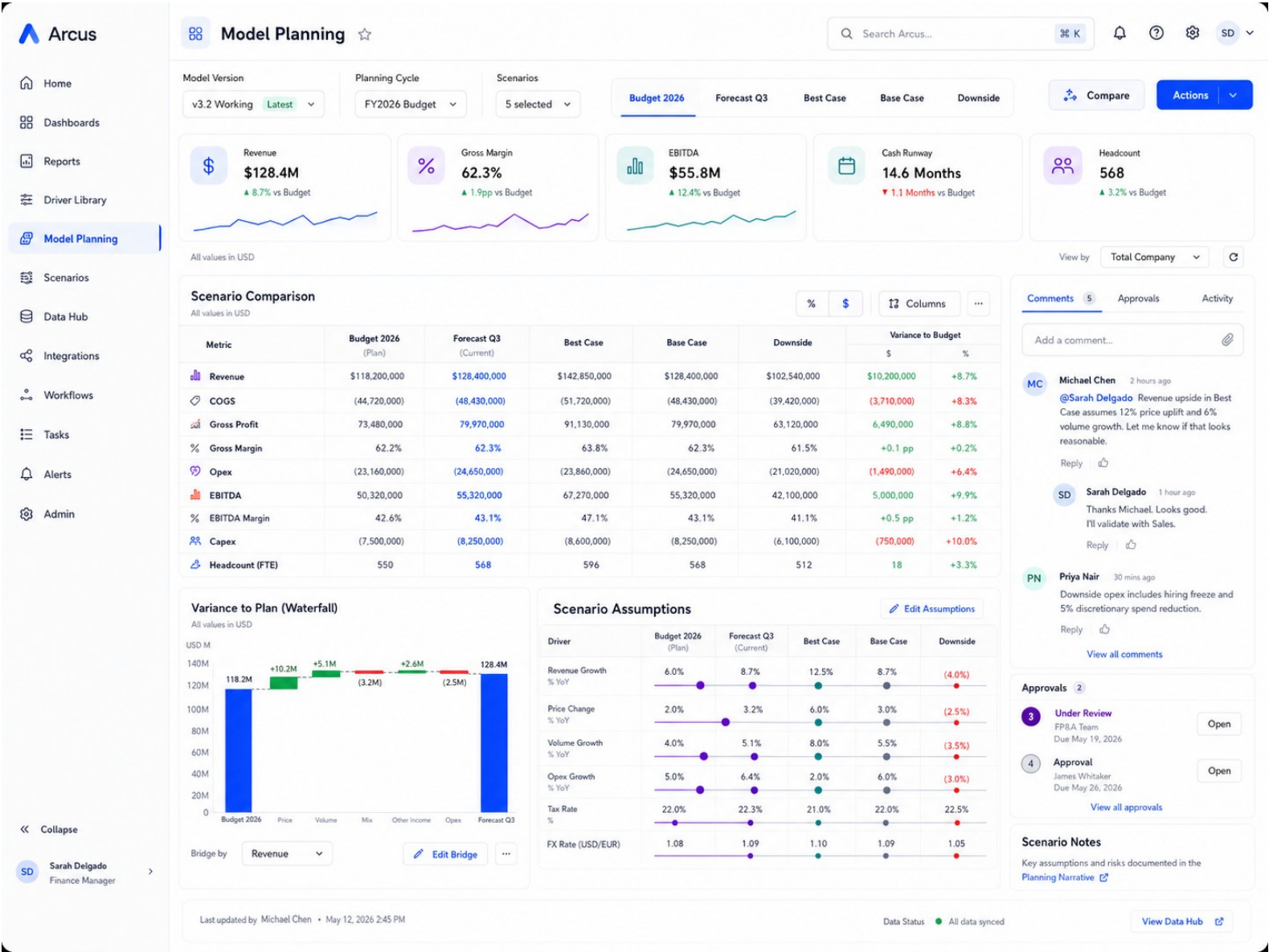
A.1 Model Planning - Planning Workspace

Planning workspace showing version and cycle selection, scenario tabs, KPI cards, planning grid, trend analysis, driver assumptions, comments, tasks and workflow status.



A.2 Model Planning - Scenario Comparison

Scenario comparison workspace showing Budget, Forecast, Best Case, Base Case and Downside metrics, variance bridge, scenario assumptions, comments and approval status.



A.3 Model Builder - Modules, Line Items and Dependency Map

Model Builder workspace showing module and dimension navigation, line-item formula grid, properties, formula editor, references and dependency map.

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SD

Sarah Delgado

Finance Manager

Model Builder

Model

FY2026 Financial Model

Workspace

Budget 2026

Valid

Circular Check Passed

Last Published: May 12, 2026 9:15 AM by Sarah Delgado

Validate Model

Publish

Modules

Search modules...

Revenue Planning

Revenue Summary

Units & Pricing

Discounts & Adjustments

Net Revenue

Revenue by Segment

Opex Planning

Workforce

Products

Regions

Versions

Time

Dimensions

Search dimensions...

Time

Monthly / Jan 2026 - Dec 2026

Product

All Products (14)

Region

All Regions (8)

Customer Segment

All Segments (4)

Version

Budget 2026, Forecast Q3, Best Case

Scenario

Budget, Upside, Downside

Currency

USD

Revenue Planning / Units & Pricing

Line Items

Templates

Validations

History

Line Item

Formula

FY2026 (USD)								
			Jan 2026	Feb 2026	Mar 2026	Apr 2026	May 2026	Jun 2026
1	Units Sold	Input	10,480	10,645	10,821	11,056	11,287	11,432
2	Price	Input	95.00	95.00	95.50	96.00	96.00	96.50
3	Revenue	= [Units Sold] * [Price]	996,560	1,011,275	1,033,006	1,060,128	1,083,552	1,103,000
4	Discount %	Input	5.0%	5.0%	5.0%	5.0%	5.0%	5.0%
5	Discounts	= [Revenue] * [Discount %]	49,828	50,564	51,650	53,032	54,178	55,164
6	Net Revenue	= [Revenue] - [Discounts]	946,732	960,711	981,356	1,007,150	1,029,374	1,048,000
7	COGS per Unit	Input	40.00	40.20	40.40	40.50	40.70	40.90
8	COGS	= [Units Sold] * [COGS per Unit]	419,200	427,938	437,204	447,768	459,777	468,689
9	Gross Profit	= [Net Revenue] - [COGS]	527,532	532,773	544,152	559,383	569,597	579,350
10	Gross Margin %	= [Gross Profit] / [Net Revenue]	55.7%	55.5%	55.5%	55.5%	55.3%	55.2%

10 line items

View: Monthly

Currency: USD

Dependency Map

View

Module

Layout

Left to Right

Show Line Items

Units & Pricing

Revenue Summary

Gross Profit Analysis

Financial Statements

Units Sold

Price

Revenue

Net Revenue

Net Revenue

Revenue by Segment

Revenue by Region

COGS

Gross Profit

Gross Margin %

Income Statement

P&L Summary

KPIs

Direct Reference

Indirect Reference

External Input

Properties

Formula Editor

References

Line Item

Revenue

Description

Total revenue before discounts and adjustments.

Format

Number

Currency

USD

Decimal Places

0

Summary Method

Sum

Dimensionality

Time

Product

Region

Customer Segment

Version

Data Type

Calculated

Formula

= [Units Sold] * [Price]

Validate

No issues

References

Units Sold

Revenue Planning / Units & Pricing

Price

Revenue Planning / Units & Pricing

Used In (2)

Created

Apr 28, 2026 by Michael Chen

Last Modified

May 12, 2026 by Sarah Delgado

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A.4 Model Builder - Formula Trace, Mapping and Validation

Detailed builder workspace showing line-item calculations, formula trace, data mappings, audit history, exceptions, validation summary and impact analysis.

ARCUS

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Sarah Delgado

Finance Manager

Model Builder

Revenue & P&L Forecast

Validation Status

Validated

as of May 12, 2026 9:12 AM

Test Calculation

Change History

Publish

Line Item Builder

View

Monthly

Show hidden

Add Line Item

Line Item	Type	Formula	Jan '26	Feb '26	Mar '26	Apr '26	May '26	Jun '26	Jul '26	Aug '26	Sep '26	Oct '26	Nov '26	Dec '26	FY2026
Starting ARR	Driver	—	125,000	126,250	127,513	128,788	130,076	131,377	132,691	134,018	135,388	136,711	138,078	139,459	1,581,319
New Bookings	Driver	—	8,500	9,100	9,800	10,400	10,900	11,300	12,000	12,400	12,900	13,300	13,700	14,200	138,500
Churn	Driver	—	(1,250)	(1,337)	(1,414)	(1,502)	(1,589)	(1,667)	(1,745)	(1,824)	(1,922)	(2,003)	(2,083)	(2,168)	(18,514)
Net ARR	Calculated	= [Starting ARR] + [New Bookings] + [Churn]	132,250	134,013	135,899	137,686	139,387	141,010	142,946	144,584	146,335	148,008	149,695	151,491	1,699,310
Payroll Expense	Calculated	= [Headcount] * [Avg Fully Burdened Cost]	3,492	3,527	3,586	3,612	3,658	3,701	3,746	3,792	3,834	3,879	3,925	3,972	44,723
Marketing Spend	Driver	—	1,250	1,300	1,350	1,400	1,450	1,500	1,600	1,650	1,700	1,750	1,800	1,900	17,550
EBITDA	Calculated	= [Net ARR] - [Payroll Expense] - [Marketing Spend]	127,508	129,186	131,013	132,674	134,279	135,809	137,600	139,142	140,801	142,379	143,970	145,619	1,637,037
Cash Balance	Calculated	= PREV[Cash Balance] + [EBITDA] - [CapEx]	52,340	105,812	160,125	212,799	267,886	323,345	380,363	439,505	500,306	563,085	627,055	692,734	692,734

Formula Trace: EBITDA

Formula

== [Net ARR] - [Payroll Expense] - [Marketing Spend]

Summary

Precedents (3)

Dependents (2)

Calculation Chain

Line Item Type

Calculated

Data Type

USD

Aggregation

Monthly

Format

#,##0

Last Calculated

May 12, 2026 9:12 AM

Status

Valid

Formula Updated

May 8, 2026 4:31 PM

by Michael Chen

Calculation Chain

Starting ARR

1,581,319

New Bookings

138,500

Churn

(18,514)

Net ARR

1,699,310

Payroll Expense

44,723

Marketing Spend

17,550

EBITDA

1,637,037

Drivers

Calculated

Result

Data Mapping

96% Mapped

Source System

All Sources

Search mappings...

Source Field

Source System

Target Line Item

Status

GL_ACCOUNT.4000

NetSuite GL

Payroll Expense

✓

GL_ACCOUNT.6100

NetSuite GL

Marketing Spend

✓

GL_ACCOUNT.8100

NetSuite GL

CapEx

✓

SALES_ORDER.AMOUNT

Salesforce

New Bookings

✓

CUSTOMER.CHURN.FLAG

Salesforce

Churn

✓

HR_HEADCOUNT

Workday HR

Headcount

✓

HR_AVG_FULLY_BURDENED_COST

Workday HR

Avg Fully Burdened Cost

✓

BANK_BALANCE

Plaid (Bank)

Opening Cash Balance

✓

GL_ACCOUNT.2000

NetSuite GL

Other Operating Expense

⚠

GL_ACCOUNT.7000

NetSuite GL

R&D Expense

⚠

Show unmapped only

View all mappings

Model Audit Log

View all

Time	User	Action	Details
May 12, 2026 9:12 AM	MC Michael Chen	Updated Formula	EBITDA
May 12, 2026 9:05 AM	SD Sarah Delgado	Updated Mapping	Payroll Expense
May 12, 2026 8:59 AM	PN Priya Nair	Added Line Item	Cash Balance
May 12, 2026 8:45 AM	JW James Whitaker	Updated Driver	Churn
May 12, 2026 8:30 AM	MC Michael Chen	Validated Module	Revenue & P&L
May 11, 2026 6:12 PM	DL Daniel Lee	Updated Formula	Payroll Expense
May 11, 2026 5:48 PM	SD Sarah Delgado	Imported Data	NetSuite GL

Exceptions

Errors 2

Warnings 2

Info 4

Type	Line Item / Mapping	Issue	Impact
⚠	R&D Expense	Missing required mapping	High
⚠	Other Operating Expense	Source field not found in source system	High
⚠	Churn	Driver has no data for Dec '26	Medium
⚠	CapEx	Negative value detected in Jun '26	Medium
⚠	Avg Fully Burdened Cost	Using fallback rate table	Low
⚠	Cash Balance	Depends on unfixed prior period	Low

View all exceptions

Validation Summary

13 Total Checks

Passed 9 69%

Warnings 3 23%

Errors 1 8%

View validation report

Impact Analysis (if Net ARR +5%)

EBITDA Impact

+84,965

+5.19%

Cash Balance Impact

+84,965

+5.19%

Base Case

Net ARR +5%

Jan '26

Mar '26

May '26

Jul '26

Sep '26

Nov '26

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